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CORE 16 SUBROUTINES 3/15/73 JBL/BPC (SUBRS,63)

/P POINTERS

BITS=0
BILLTT=6
SUTD=7
PROGNM=11
DDTSEG=16
RESTPC=32
DDTBKP=20
HH16PC=41
HH16IO=42
USERAC=44
USERPC=45
USERIO=46
USERFG=47
C16TEM=55
DDTS6=65
IOPMAX=74
TISMAX=75
TTNO=76
TRAPPC=77
ERCODE=102
OWNWD=104
TTTSU=105
IOPTSU=106

/TEM STORAGE IN CORE 0

FSA=107
STS=110
JMODE=111
ATEM=112
FAC=123
FOV=125
BTEM=127 /ALSO REFERENCED AS CBTM3 (BTEM+3)
DTEM=150
ETEM=373

FILE BEG = 127

10/1/86
Completed with
TV, JBL, ...

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SZS=640000
DELAY=IOT 1600
EOTHNG=IOT 101
TYI=IOT 201
TYO=IOT 301
TIS=IOT 401
TOS=IOT 501
GTY=IOT 600
CTTS=IOT 1001
ADDLP=IOT 1300
WPP=IOT 3200
GTD=IOT 3400
PEEK=IOT 3700
C16RET=IOT 4000
RPA=IOT 4200
PPA=IOT 4500
RCK=IOT 4700
RPPU=IOT 5501
RPPB=IOT 5541
EXPP=IOT 5741
BEDTR=IOT 5701
RLSQTk=IOT 6041
RAI=IOT 6101
RAB=IOT 6141
EXAI=IOT 6201
EXAB=IOT 6241
WNBf=IOT 6341
WNI=IOT 6301
WNIH=IOT 6321
WAI=IOT 6401
WAIH=IOT 6401
WAIA=IOT 6411
IVNR=IOT 6500
WNIA=IOT 6701
HALT.=IOT 12401

/TEM STORAGE IN CORE 16

A=0
B=1
C=2
D=3
E=4
F=5
G=6
H=7

/EXECUTIVE FUNCTIONS

```

160010/  C16RET+1      /ILLEGAL

PSTART,  LIO C6
          JMP START1
STARTUP,  LIO C14      /HALT IF OVF SET
          SZO I
          LIO (15)     /STARTUP
START1,   DZM I (IOPMAX)
          LAW 36
          IVNR 23
          JMP .
          C16RET      /DEBREAK VIA C16PC IN ITEM READ

HHORR,    LAW DDTSEG
          WPP          /SAVE TRPBUF
          GTD+1
          LAW DDTBKP+7
          WPP          /SAVE TIME AND DATE OF CRASH IN ...
          GTD+1      /... LOWER CORE
          LIA
          LAW DDTBKP+6
          WPP
          CLA
          WNBFF+2     /WRITE OUT FIRST 50. WORDS AS A BLOCK
          JMP .
          LAW 37      /PUT DRA OF BLOCK IN REWRITE NUMBER ...
          WPP        /... OF ITEM TO BE WRITTEN
          LAW 36
          LIO (I-36)  /LENGTH
          WPP
          WNI+2
          -JMP .
          LAW DDTBKP
          WPP
          LAW DDTSEG
          PEEK
          C16RET+3    /SOROBAN TYPEOUT
          LAC (HHORRT)
          TOS
          HLT
          HALT.

```

```

HHORRT:  TEXT /
PROGRAM CANNOT CONTINUE.#/

```

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/START UP A NEW USER PROGRAM.

SUPGO, LIO I (USERAC)
LAW RESTPC
WPP
LIO I (USERIO)
LAW 36
DZM I (IOPMAX)
RAI+2
JMP .
DIO I (USERIO)
LAW RESTPC
PEEK
DAC I (USERAC)
DZM I (DDTS6+1)
SPA
LAW DDTS6+1
DAC I (USERPC)
LAW BILLTT
PEEK
DAC I (TTNO)
C16RET

/GET A CLEAN FILE PACKAGE.

FILPKG, DZM I (IOPMAX)

LIO FILIVN

LAW FILBEG

IVNR+23

JMP .

LAC FILCAL

DAC I C101

JMP R1

✓ FILIVN,
FILCAL,

251

JMP FILBEG+15

/use low core (127-637)

/use CAL instruction

/CORE 16 BREAKPOINT LOGIC

```

BKPNT,    LAC (177777)
          AND I (HH16PC)
          SAS (DDTS6+1)
          SAD (DDTS6+2)
          JMP GETDDT    /DDT IS EXECUTING AN INSTRUCTION
          RAL 2S
          SPA          /FROM CORE 16?
          JMP GETDDT    /YES, GET DDT
          LAW 7777
          AND I (HH16PC)
          DAC A        /ADDRESS OF INSTRUCTION
          LAW DDTBKP
          DAC B        /ADDRESS OF BREAKPOINT(BPT)
16BPTL,   PEEK        /READ FIRST BPT INTO AC
          DAC C        /CONTENTS OF BPT REGISTER
          AND (7777)   /?
CHARLH,   SAD A        /IS THIS THE BPT?
          JMP 16BPTF    /YES
          IDX B        /NO, LOOP
          IDX B
          SAS (DDTBKP+10) /NOT A BPT?
          JMP 16BPTL    /NO
          JMP GETDDT

16BPTF,   IDX B
          PEEK        /GET PROCEED COUNTER INTO AC
          LIO B
          ADD B17      /IDX PROCEED COUNTER
          SWP
          WPP        /WRITE OUT PROCEED COUNTER
          SPI I
          JMP GETDDT
          LAC I A
          XOR C        /PUT ADDRESS INTO REG.
          AND (17777)
          XOR C
          DAC I (DDTS6+3)
          LAW 1
BØ,       ADD A        /GET PC
          ADD CJ        /?
          DAC I (DDTS6+4)
          IDA
          DAC I (DDTS6+5)
          LAC BØ
          AND I (HH16PC) /OVF BIT
CIORIN,   IOR (DDTS6+3)
          DAC I (HH16PC)
          C16RET+2     /DEBREAK THRU HH16

```

/DDT-GO LOGIC. DDT'S IOPMAX IS ZERO. IOT'S OVF AND LINK CLEAR.
/REWRITES ON HELD DONE WITH "DON'T CHECK OWNWD OR REWRITE NUMBER".

```
DDTGO,    LAW DDTSEG      /OVF, LINK CLEAR
          PEEK+20
          LAW 36
          WAIH+22        /WRITE DDT OUT
          JMP DDTWER
          SZF 1
          JMP DDTUF      /"U"F", "C"F", OR "L"F"
          LIO (DDTSEG+1  /READ CORE IMAGE
          PEEK+60
          RAI+2
          JMP DDTERR
          SZF 4
          JMP DDTZ       /"Z"
          SZF 5
          JMP DDTSF      /"S"F"
          SZF I 6
          C16RET+2       /REGULAR "GO"
          LIO A          /P"F". LENGTH OF ITEM APPEARED IN A.
          WPP
          LIO I (HH16IO  /USER PUT DRA IN REG 42 OF CORE IMAGE
```

/SAVE NEXT 2 INSTRUCTIONS WHEN WAIH+22 WILL REWRITE ITEM ON FREE,
/AS LONG AS REWRITE NUMBER AND OWNWD REALLY DO MATCH UP.

```
          WAI+2          /REWRITE ITEM (PRESUMABLY ON LIBRARY)
          JMP DDTERR
          JMP DDTPFX
```

```
DDTUF,    LIO I (USERIO  /DDT'S I, O. HAS DRA OF FILE OR...
          RAI+2          /...OF STARTUP PGM.
          JMP DDTERR
          SZF 2
          JMP DDTCAL     /"C"F" OR "L"F"
GETDDT,   CLO           /BPT LOGIC MAY HAVE SET OVFL
BKKEY,    LAW 36        /BK KEY ENTERS HERE WITH OVF SET
          LIO (I-36
          WPP
          LIO (DDTSEG+1
          PEEK+60
          WAIH+22        /WRITE OUT CORE IMAGE
          JMP DDTERR
```

```
DDTPFX,   DDTSFX,     LAW 36      /ENTRY FROM DDTERR, DDTUF, DDTSF
          LIO (DDTSEG
          PEEK+60
          DZM I (IOPMAX
          RAI+2        /READ DDT IN
          JMP .         /CAN'T FIND DDT
          IDX I (USERPC
          SZL
          C16RET       /ERROR RETURN
          SZO
          JMP R1        /BK KEY
          JMP R2        /ILLEGAL INSTR OR BPT OR NORMAL RETURN
```

```

DDTCAL,    LAW 100           /SA OF STARTUP PGM SEG
           DAC I (HH16PC
           LIF
           DIO I C101        /C"F" VS L"F" DISTINCTION FOR SUP SEG
           C16RET+2

DDTZ,      LAW RESTPC
           PEEK+20           /GET G# INTO I. O.

CLACB,
DDTZL,     LAC B             /UPPER LIMIT APPEARED IN B
           SUB A             /LOWER LIMIT IN A
           SPA
           JMP GETDDT
           DIO I A
           IDX A
           JMP DDTZL

DDTSF,     WNIH+2           /WRITE COPY OF CORE
           JMP DDterr
           LAW DDTSEG+1
           WPP               /RETURN DRA OF SAVED FILE IN DDTSEG+1
           JMP DDTSFX

DDterr,    LAW RESTPC       /IOP ERROR
           LIO I (ERCODE
           WPP               /SAVE ERCODE FOR DDT IN RESTPC
           CLL"U"CML
           JMP DDTSFX        /READ DDT AND GIVE R1

DDTWER,    LAW RESTPC       /ERROR TRYING TO WRITE DDT OUT
           LIO I (ERCODE
           WPP               /SAVE ERCODE WHERE DDT WILL BE LOOKING FOR IT
           JMP R1            /GIVE ERROR RETURN, DDT STILL IN CORE.

/FASTRAND SWAPPING
C16FIN,    C16RET+4
C16FOU,    C16RET+5
C16FSW,    C16RET+6
           C16RET+10
C16SWF,    C16RET+7

LISSUP:    ADDLP 2
           C16RET

16LSSP:    ADDLP 2
           LAC B
ML2,       JMP A

```

/USER FUNCTIONS, IOT'S

/EDIT IOT AND SUBR

EDIT,	LAW IR1	/SETUP RETURN TO GIVE RETURN TO USER
.EDIT,	LIO I (USERAC)	/SUBROUTINE ENTRY, USED ONLY IN BCJBH
.EDITJ,	DAP H	/JOB HUNTER ENTRY. SET RETURN
	DIO G	/TEXT PTR FOR EDITING
	DIO F	
	DIO E	/BACKSPACE STOPPER
	LCH I E	/JUST TO IDC IT
EDITL,	LCH I F	
	SAD ML4	/EOM?
	JMP EDITE	
	SAS ML6	
	JMP EDITS	/NOT A WARNING CHAR.
	LCH I F	/GET 2ND HALF
	SAD CHARLD	/(440000) BS?
	JMP EDITB	
	SAD ML4	/RO?
	JMP EDITR	
	RAR 6S	
	IOR ML6	/SETUP TO DCH TWICE
	DCH I G	
EDITS,	DCH I G	/COPY BACK INFO
	JMP EDITL	

/BACKSLASH

CLACG,

EDITB,

LAC G	
IDC	
SAD E	
JMP EDITL	/CAN'T BACK UP
LAC G	
SMA	
SUB B1	
SUB B1	/UNSTEP
DAC G	
LCH G	
SAS ML6	
JMP EDITL	
JMP EDITB	/PREV. CHAR, WAS A WARNING CHAR.

/EOM

EDITE,

EDITR,

IDX H	/SECOND EXIT
LAC ML4	
DCH I G	/STORE EOM
LIO G	/RETURN PTR TO JBH IN IO
SZF I 1	/AND TO USER IN AC EXCEPT ON EDIT+40
DIO I (USERAC)	
JMP HEXIT	/GO EXIT THRU H

/RPB SUBROUTINE AND IOT

```

RPB,      LAW R1
.RPB,     DAC H           /EXIT
          LAW I 3
          DAC G           /CTR FOR RPA'S
RPB1,     RPA
          RIR 8S
          SPI I           /CHANNEL 8 BIT?
          JMP RPB1        /NO, IGNORE CHAR.
          RIL 2S
          LAC I (USERIO)
          RCL 6S
          DAC I (USERIO) /SHIFT INTO USER'S IO
          ISP G
          JMP RPB1        /3 TIMES
HEXIT,    LAC CJ
CDIPH,    DIP H
CJMPH,    JMP H           /EXIT THRU H

```

/ENTER READIN MODE (ERIM) IOT

```

ERIM,     DZM F           /USERCORE PTR. FOR DIO'ING
ERIM1,    JSP .RPB        /USER MUST HAVE READER
          LAC I (USERIO)
          SPA
          JMP GO          /GO TO LOC. SPECIFIED IN AC
          DAP F
          JSP .RPB
          LAC I (USERIO)
          DAC I F
          JMP ERIM1       /LOOP UNTIL TERMINATOR

```

/PPB IOT

```

PPB,      LAW I 3         /PUNCHES 3 LINES OF TAPE
          DAC G
          LIO I (USERIO)
PPB1,     LAW 2           /8TH HOLE
          RCL 6S
          SWP
          PPA
          SWP
          ISP G
          JMP PPB1
          JMP R1

```

/TTON IOT

TTON, GTY /IO POSITIVE
 DIO I (USERIO)
 SPI
 JMP R1 /CAN'T GET TT
 LAC (TTONT)
 TOS"U"1 /TURN ON TT AND SEND WRU'S
 JMP TTONR2
 RCK
 DAC H

TTONL, CLA"U"CLI
 DELAY+40
 RCK
 SUB H
 SPA
 ADD C60,K
 SUB (17777) /8 SECONDS
 SMA
 JMP R1 /WON'T TURN ON
 TYI"U"3 /TEST FOR "HERE IS"
 JMP TTONR2
 SPI
 JMP TTONL

TTONR2, LAC (760002) /ACTS LIKE LAW 3 WITH DELAY+40
 DELAY+40
 TYO"U"1
 NOP
 JMP R2

TTONT: 770077
 007700
 774177
 417741
 770077
 007700
 774177
 417741
 774177
 417705
 770574

/HALT IOT
HALT,

CTTS
NOP
DZM I (JMODE)
LAC (760000)
TYO
JMP HALT1 /TRY ONCE MORE
GTD /OLD TIME
DAC A
DIO B
GTD+1 /CURRENT TIME
SAD A
STF 3 /SUPPRESS DATE IF SAME
DAC I (USERAC)
DIO I (USERIO)
LAC CHLT
DAC I (TTTSU)

HALT1,

JSP .STD
LAW BITS
PEEK
SPA /RUNNING UNDER IDDT?
JMP I (66) /YES, SILENT RETURN
LAC (HALTT)
TOS

CHLT,

HLT /GIVE UP
LAW 37 /SEE IF SHOULD TYPE EOT
SUB I (TTNO)
SPA
JMP HALT2
SUB (37-30)
SPQ

HALT2,

HLT
LAC (HLTEOT)
TOS
JMP SICKTT
HLT

HALTT: 771476

767676

767674

HLTEOT: 770474

/DECODE NUMBER IOT AND SUBROUTINE

```

DNM,      LAW TR1      /USER ENTRY
.DNM,     DAP H        /SUBR ENTRY
.DNM+1,   CLO
          LAC I (FSA)   /TEXT PTR.
          DAC G
          LAC I (USERAC) /SAVE UP MAX IN ATEM
          DAC I (ATEM)
          DZM I (USERAC) /INIT. FOR NUMBER
          LCH I G
          SAD CHLPLS     /CHECK 1ST CHAR FOR SIGN
          JMP DNMPPL
          SAD (CHARAC L-)
          JMP DNMM
DNMDO,    SZF 5         /OCTAL?
          JMP DNMOCT
DNMDI,    RAL 6S
          XOR B13        /CONVERT INTERNAL TO OCTAL
          DAC I (ATEM+1)
          SUB C10,
          SMA
          JMP DNMDDE     /NON-NUMERIC
          STF 1          /SET NUMERIC FLAG
          LAW 10,        /RADIX
          MUL I (USERAC)
          DIV B17
          JMP DNMMO      /OVERFLOW
          ADD I (ATEM+1)
          DAC I (USERAC)
DNMDL,    LCH I G      /NEXT CHAR.
          JMP DNMDI

```

/MINUS SIGN

```

DNMM,     SZF 4
          JMP DNMIILL    /- WAS ILLEGAL
          STF 2          /REMEMBER SIGN
          JMP DNMISS     /SKIP SPACES

```

/PLUS SIGN

```

DNMPPL,   SZF 3
          JMP DNMIILL    /+ WAS ILLEGAL
DNMISS,   LCH I G      /SKIP SPACES
          SZA I
          JMP DNMISS
          JMP DNMDO      /TEST RADIX

```

/OVERFLOW

```

DNMMO,    LAC B0
          ADD B0         /SET OVF FF
          JMP DNMDL

```

/ILLEGAL FORMAT

```

DNMIILL,  DZM I (USERAC)
          JMP DNMUX1     /DON'T CHANGE FSA, R1

```

/NON-NUMERIC CHARACTER ON DECIMAL DECODE

```

DNMDE,      SZF I 1
            JMP DNMill      /NO NUMBER, ILLEGAL
            LIO I (USERAC)
            SZF I 6
            JMP DNMDSM      /SKIP MAX. TEST
            LAI              /GET ABS. VAL OF NUMBER
            SUB I (ATEM)     /SUB MAX
            SUB B0           /SET OVFF FF IF AC IS 0 OR MINUS
DNMDSM,     SZF 2
ML6,        CMI              /COMP. IF NEGATIVE
            DIO I (USERAC)
DNMOSM,     SZO
            JMP DNMOVFF
            IDX H            /OK R2
DNMUX,      LAC G            /UNSTEP
            SMA
            SUB B1
            SUB B1
            DAC I (FSA)
DNMUX1,     CLF 1
            CLF 2
            JMP HEXIT       /EXIT THRU H

```

/OCTAL DECODE LOOP

```

DNMOCT,     CLI"U"SWP      /CHAR. INTO IO, CLA
            RCL 3S
            SAS B16         /IS CHAR. "2X" IN INTERNAL
            JMP DNMOE       /NO, TERMINATOR
            STF 1
            LAC I (USERAC)
            RCL 3S          /SHIFT IN NUMBER
            DAC I (USERAC)
            LAI              /OVERFLOW TO AC
            SUB B0          /SET OVFF FF IF POS. QUANT.
            LCH I G
            JMP DNMOCT

```

/NON-NUMERIC CHAR. IN OCTAL DECODE

```

DNMOE,      SZF I 1
            JMP DNMill
            LAC I (USERAC)
            SZF 2
            CMA              /COMP. IF NEGATIVE
            DAC I (USERAC)
            SZF 6
            SZA I
            JMP DNMOsm      /SKIP MAX. TEST
            SCM"U"IDA       /2'S COMP. CMA
            TAD I (ATEM)     /LEAVE LINK FF=0 IF OVERFLOW
            SZL
            JMP DNMOsm      /OK

```

/OVERFLOW ERROR

```

DNMOVFF,    CLC              /AC GETS -0
            DAC I (USERAC)
            JMP DNMUX       /R1, FSA UPDATED

```

/COUNT DIGITS (COUNT IN F) SUBROUTINE

CDG,	DAP H	
	LAC I (FSA)	
	DAC G	
CDGL,	LCH I G	
	RAL 6S	
	XOR B13	
	SUB C10,	
	SMA	
	JMP .DNM+1	/NON-NUMERIC, OK GO DNM
	ISP F	/COUNT NUMERIC CHAR.S
	JMP CDGL	/LOOP
	JMP HEXIT	/R1, TOO MANY CHAR'S

/SUBR TO TYPE OR SETUP FROM IO, USES PF6, ROTATES IO

```

TYOIF,      DAP H
            LAI
            SZF 6
            JMP TYOIF1      /TYPE
            DCH I G
            JMP TYOIF2
TYOIF1,     TYO
            JMP SICKTT
            RAL 6S
TYOIF2,     LIA
            JMP HEXIT

```

/SETUP NUMBER SUBR AND IOT

```

SNM,        JSP STYOIF      /SKIP TYPING FOR JBH MAYBE
.SNM,       LIO I (USERAC) /GENERAL-PURPOSE SUBR ENTRY
            /ABOVE USED ONLY BY BC JBH?
.SNMJ,      /JOB HUNTER ENTRY
            DIO I (ATEM+5) /OUTER LOOP TERMINATOR
            DAP I (ATEM+2) /RETURN
            LAC I (STS)
            SZF I 6        /TYPE BIT
            DAC G
            LAC I (USERIO) /RIGHT ADJUST COUNT
            IDA"U"LIF
            DIO I (ATEM+6)
            CMA"U"CLI
            DAC I (ATEM)
            LAW 10.
            SZF 5
            LAW 10        /OCTAL
            DAC I (ATEM+1) /RADIX
            LAC I (ATEM+5) /GET NO. BACK
            SZF I 5        /NO SIGN FOR OCTAL NUMBERS
            SMA"U"SZF 3    /INHIBIT POSITIVE SIGN BIT
            JMP SNMNS      /SKIP SIGN TYPEOUT
            SZF 4
            LIO CHLPLS
            SPA
            LIO (CHARAC L-)
            SPA
            CMA            /ABS. VAL
            DAC I (ATEM+5) /RE-DAC OUTER LOOP TERMINATOR
            IDX I (ATEM)
            JSP TYOIF      /TYPE SIGN

```

```

SNMNS,   DZM I (ATEM+4) /INNER LOOP TERMINATOR
SNMOL,   LAC I (ATEM+5)
SNML,    DAC I (ATEM+3) /INNER LOOP VALUE
          CLI"U"SWP
          RCL 1S
          DIV I (ATEM+1)
C60.K,   60000,
          SAS I (ATEM+4)
          JMP SNML      /KEEP DIVIDING (INNER LOOP
          LAW 20
          AAI           /CHAR. MADE IN AC
          RCR 6S
          IDX I (ATEM)  /CHAR COUNT
          SZF I 1       /RIGHT ADJUST BIT
          JSP TYOIF     /TYPE CHAR. IF NOT RIGHT ADJUST
          LAC I (ATEM+3)
          DAC I (ATEM+4) /RE-INIT INNER LOOP
          SAS I (ATEM+5)
          JMP SNMOL     /GET NEXT CHAR. (OUTER LOOP)
          SZF 1
          JMP SNMRA     /RIGHT ADJUST
          LAC G
          SZF I 6       /TYPE BIT
          DAC I (STS)
          LAC ML4       /(CHARAC L#)
          SZF I 6
          DCH I G       /DCH EOM AFTER PTR.
          LIO I (ATEM+6)
          LFI
          LAC I (ATEM+2)
          DAP H
          JMP H

SNMRAL,  SZF 2         /LEADING ZEROES BIT
          LIO B1       /CHARAC L0
          JSP TYOIF
SNMRA,   CLI"U"CLF 1   /RIGHT ADJUSTER
          ISP I (ATEM)
          JMP SNMRAL
          JMP SNMNS     /DO WHOLE THING OVER AGAIN, LESS SIGN

```

```

/ROUTINE TO GIVE IMMEDIATE RETURN 1 IF SETUP ROUTINE
/IS GOING TO TYPE AND JMODE IS NEGATIVE, PRECEDES ALL SETUP ROUTINES.
STYOIF,  DAP H
          LAC I (JMODE)
          SPA"U"SZF 6

```

```

CJR1,    JMP R1
          LAC CJR1     /RETURN WITH R1 IN ADDRESS OF AC
          DIP H
          JMP H

```


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/SUBR TO TYOIF, MAKING USE OF DISCREPANCY BETWEEN G AND I (STS)
TYOIFR, DAP H
LAI
SZF 6
JMP TYOIF1
DCH G
LAC G
DAC I (STS)
JMP HEXIT

/TIME AND DATE ROUTINES

/SUBR TO SETUP DATE FROM AC INTO D-E-F

```

.DNUM,   DAP H
          LAC I (USERAC) /DAYS SINCE 1/1/1849
          SUB (18686.) /DAY # OF 3/1/1900
          SMA
          IDA /DISPLACE DAYS AFTER 3/1/1900 UP 1
          ADD (18686.)
          CLI"U"SWP
          RCL 1S
          DIV C1461. /# OF DAYS IN A 4-YEAR CYCLE
C1461., 1461.
          SAL 2S
          DAC F /YEARS IN INTEGRAL 4-YEAR CYCLES
          LAI
          SUB C1154. /DAY # OF 2/29 OF FOURTH YR OF CYCLE
          SZA I
          JMP SUTD29 /HANDLE 2/29 SEPARATELY
          SMA
          SUB B17 /DISPLACE DAYS AFTER 2/29 DOWN 1
          ADD C1154. /NOW PRETEND ALL YEARS HAVE 365, DAYS
          CLI"U"SWP
          RCL 1S
          DIV C365. /GET # OF YEAR WITHIN CYCLE
C1154., 1154.
          ADD F /ADD YEARS IN 4-YEAR CYCLES
          ADD C1849. /AD IN ORIGIN OF YEARS
          DAC F /EXACT YR #
          DIO D /# OF DAY WITHIN YEAR
          LAC (SUB SUTDTB-1) /INIT TO FIND MONTH
          DAC B
          LAI
SUTDML, DAC E /STORE IN LOOP, ANSWER ON IEXIT
          IDX B /LOOK AT NEXT MONTH
          LAC D
          XCT B /SUBTRACT FROM TABLE
          SMA
          JMP SUTDML /CONTINUE UNTIL NEGATIVE
          IDX E /EXACT DAY OF MONTH
          LAC B
          SUB (SUB SUTDTB-1) /CALCULATE MONTH
SUTDM, DAC D /STORE MONTH
          JMP HEXIT

```

/SUBR TO SETUP TIME FROM IO INTO B-C

```

.TNUM,   DAP H
          CLA
          LIO T (USERIO) /MINUTES SINCE MIDNIGHT
          RCL 1S
          DIV (60.)
C2020, 2020
          DAC B /HOURS
          DIO C /MINUTES
          JMP HEXIT

```

/FEB 29 CASE

SUTD29, LAW 1852.
 ADD F
 DAC F
 LAW 29.
 DAC E
 LAW 2
 JMP SUTDM

/ADD YEAR ORIGIN + 3
 /TO # OF YEARS FROM 1 YEAR CYCLES
 /DAY = 29.

/TABLE OF # OF DAYS IN A YEAR PREVIOUS TO FIRST OF ENTRY MONTH

CZ, 0
 SUTDTB, 31.
 59.
 CBTM3, 90.
 CJOBP, 120.
 151.
 181.
 212.
 243.
 273.
 304.
 334.
 C365., 365.

/ARGUMENT INDIRECT TRACING ROUTINES

TRACEI, DAP H
 CHLCOL, DIO A
 JMP TRACEL

TRACE, DAP H /ENTRY TO TRACE WORD AFTER IOT
 IDX I (USERPC)

DAC A
 TRACEL, LAC I A
 LIA
 AND (607777)
 DAC A
 RIL 5S
 SPI
 JMP TRACEL
 JMP HEXIT

/LOOP UNTIL I BIT SET
 /PTR IN A

/IOT TO CONVERT FROM 2 WD TO 5 WD FORMAT

TDNUM, JSP TRACE
 JSP .DNUM
 JSP .TNUM
 LAC CLACB
 DAC G
 TDNUML, XCT G
 DAC I A
 IDX A
 IDX G
 SAS CLACG
 JMP TDNUML
 JMP R1

/INIT TO COPY INTO USER CORE

/SETUP TIME AND DATE, IOT AND SUBR

```

STD,      JSP STYOIF
.STD,     DAP A          /EXIT THRU A
          LAC I (USERAC
          DAC I (ATEM+7   /SAVE UP AC
          LAC I (USERIO
          DAC I (ATEM+10) /SAVE UP IO
          LIF
          DIO D          /SAVE UP FLAGS
          DIO C
          SZF 2
          JMP SUTDD      /SUPPRESS TIME BIT
          STF 3          /INHIBIT SIGN BIT IN .SNM
          CLF 5          /OCTAL BIT CLEARED IN .SNM
          JSP .TNUM      /PUT TIME IN B-C
          LAW I 12.
          ADD B
          DAC B          /SIGN BIT OF B IS AM-PM FLAG
          SPA
          ADD C14
          SZA I
          LAW 12.        /GET NUMBER FROM 1 TO 12, FOR HOUR
          DAC I (USERAC
          LAW 2          /DIGIT COUNT IF COLUMNATING
          DAC I (USERIO
          JSP .SNM
          SZF 4          /CONDENSED TIME BIT
          JMP SUTDCT
          LIO CHLCOL     /:
          JSP TYOIFR
          LAC C
          DAC I (USERAC  /# OF MINUTES
          STF 2          /LEADING ZEROS
          STF 1          /RIGHT ADJUST, 2 IS IN IO
          JSP .SNM
          LAC I (STS
          DAC G          /UNSTEP G
          LIO (FLEXO PM
          LAC B
          SPA
          LIO (FLEXO AM
          JSP TYOIF
          JSP TYOIF
SUTDTI,   JSP TYOIF
          LIO G
          LAC ML4
          SZF I 6
          DCH I G
          DIO G
          LIO D          /RESTORE FLAGS
          DIO C
          LFI"U"SCI
          SZF I 3        /SUPPRESS DATE BIT
          JSP TYOIF      /TYPE SPACE IF BOTH TIME AND DATE SETUP
          LAC G          /UPDATE PTR.
          SZF I 6        /TYPE BIT
          DAC I (STS     /ONLY IF NOT TYPE+ING
          SZF 3          /SUPPRESS DATE BIT

```

JMP SUTDX
LAC I (ATEM+7 /RESTORE DATE TO AC
DAC I (USERAC

```

SUTDD,   JSP .DNUM      /SETUP DAY IN D-E-F
          LAW 2          /DIGIT COUNT
          DAC I (USERIO
          CLF 2          /LEADING SPACES
          CLF 5          /DECIMAL
          STF 3          /INHIBIT SIGN
          LAC D          /DAY WITHIN MONTH
          DAC I (USERAC
          JSP .SNM
          LIO (CHARAC L/
          JSP TYOIFR
          LAC E          /MONTH
          DAC I (USERAC
          STF 2          /LEADING ZEROS
          JSP .SNM
          LAW 2          /TEST FLAG 5
          AND C
          SZA
          JMP SUTDX      /COMPRESS DATE, NO YEAR
          LIO (CHARAC L/
          JSP TYOIFR
          LAC F          /YEAR
          DAC I (USERAC
          JSP .SNM      /OVERRIDE 2 DIGIT COLUMNATION WITH 4 DIGIT YEAR
SUTDX,   LAC I (ATEM+7)
          DAC I (USERAC /RESTORE AC AND IO
          LAC I (ATEM+10)
          DAC I (USERIO
AEXIT,   LAC CJ
          DIP A
CJ,      JMP A

/CONDENSED TIME HANDLER
SUTDCT,  LAC I (STS
          DAC G
          LIO CJ        /(CHARAC LP
          LAC B
          SPA
          LIO CHARLA     /(CHARAC LA
          JMP SUTDTI     /CONTINUE IN MAIN CODING

```

/DECODE TIME IOT, FLAGS 3,4,6

```

DTM,      LAC I (FSA
          DAC A           /SAVE FOR ERRORS
          DAC G
          LCH I G
          SAD (CHARAC LT)
          JMP DTMT        /RIGHT NOW
          LAW 12.         /MAX. FOR HOUR
          DAC I (USERAC
          LAW I 3
          DAC F
          JSP CDG
          JMP DUNILL
          LAC I (USERAC
          SZA I
          JMP DUNILL      /0 IS ILLEGAL HOUR
          SAD C14
          CLA
          MUL (60.)
          RCR 1
          DIO B
          LCH G
          SAS CHLCOL      /CHARAC L:
          JMP DTMB        /NO COLON, MUST BE IN BRIEF FORMAT
          LAC G
          DAC I (FSA      /UPDATE PTR,
          LAW I 3
          DAC F
          LAW 59.         /MAX FOR MINUTE
          DAC I (USERAC)
          JSP CDG
          JMP DUNILL
          LAW I 1
          SAS F           /REQUIRE EXACTLY 2 DIGITS
          JMP DUNILL
          LAC B
          ADD I (USERAC   /GET MINUTES SINCE MIDNIGHT
          DAC B
          LCH G
DTMB,     SZA I
          LCH I G        /PERMIT 1 SPACE
          LIO C720.
          SAD CJ         /(CHARAC LP
          JMP DTMPM      /PM, GO ADD IN 720. FROM IO
          SAS CHARLA     /(CHARAC LA
          JMP DUNILL
          CLI            /AM, ADD IN 0 FROM IO
DTMPM,    LAC B
          AAI
          DAC I (USERAC   /RETURN TIME TO AC
          LIO G           /PTR PAST A OR P
          LCH I G
          SAD (CHARAC LM
DTM2,     LIO G          /IF AM OR PM, UPDATE PTR, IN IO
          DIO I (FSA
          JMP R2

```

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/RIGHT NOW (LOWER CORE
DTMT, LAW SUTD+1
 PEEK
 DAC I (USERAC)
 JMP DTM2

/DECODE DATE IOT, FLAGS 3,4,6

```

DDT,      LAC I (FSA
          DAC A          /SAVE FOR ERRORS
          DAC G
          LCH I G        /FIRST CHARAC
          SAD (CHARAC LT)
          JMP DDTT        /N DAYS HENCE
          SAD CHARLY      /(CHARAC LY
          JMP DDTY        /N DAYS AGO
          LAW 12.
          DAC I (USERAC   /MAX FOR MONTH
          LAW I 3
          DAC F           /2 DIGITS
          JSP CDG
          JMP DUNILL
          LAC I (USERAC   /MONTH
          DAC I (ATEM+2   /SAVE FOR LATER
          LCH G
          SAS (CHARAC L/  /REQUIRE /
          JMP DUNILL
          LAC G
          DAC I (FSA      /STEP PTR.
          LAW 31.
          DAC I (USERAC   /MAX FOR DAY
          LAW I 3
          DAC F           /2 DIGITS
          JSP CDG
          JMP DUNILL
          LAC I (USERAC
          DAC I (ATEM+3   /SAVE DAY FOR LATER
          LCH G
          SAS (CHARAC L/)
          JMP DDTCY       /NO YEAR, PRESUME CURRENT
          LAC G
          DAC I (FSA      /STEP PTR.
          LAW 2099.
          DAC I (USERAC
          LAW I 5
          DAC F           /4 DIGIT MAX
          JSP CDG
          JMP DUNILL
          IDX F
          SZA I
          JMP DDTCS       /4 DIGITS, CENTURY SPECIFIED
          SAS (-2
          JMP DUNILL
          LAW 1900.       /2DIGITS, ADD IN CURRENT CENTURY
DDTCS,    ADD I (USERAC
DDTGY,    DAC I (ATEM+4) /SAVE YEAR
DCDIDI,   CLL"U"CML"U"SCI /SET LINK FOR LATER
          LAC I (ATEM+2   /MONTH
          ADD (ADD SUTDTB-2 /TABLE LOOK-UP
          DAC B

```

```

SAD (ADD SUTDTB
CLL /CLEAR LINK IF FEBRUARY
LAW I 1
ADD I (ATEM+3 /GET DAY #
XCT B /ADD IN DAYS FROM PREVIOUS MONTHS
DAC B /DAY # (0-364.)
LAW I 1849. / - ORIGIN OF YEARS
ADD I (ATEM+4 /GET YEARS SINCE ORIGIN
SCR 2S /GET # OF 4 YEAR CYCLES
RIL 2S
DIO C /GET # OF LEFTOVER YEARS IN IO
MUL C1461. /GET # OF DAYS DUE TO 4 YEAR CYCLES
RCR 1
DIO D
LAW 365.
MUL C /GET # OF REMAINING DAYS FROM LEFTOVER YEARS
RCR 1
LAI
ADD B /ADD POSITION OF DAT WITHIN YEAR
SUB C1154. /DAY 3 OF 2/28/4TH YEAR
SMA /DON'T ADD IF BEFORE 2/28/4TH YEAR
TAD CZ /ADD 1 IF NOT FEB (LINK SET)
ADD D
SUB (18686.-1154.) /ADD 1154, SUB DAY#(3/1/1900)
SMA
SUB B17 /SUB 1 IF 3/1/1900 OR AFTER
ADD (18686. /RESTORE DAT #
DAC I (USERAC)
JSP .DNUM /NOW CONVERT BACK TO MONTH/DAT/YEAR
LAC I (ATEM+2
SAS D /ALL 3 MUST COMPARE
JMP DUNILL
LAC I (ATEM+3
SAS E
JMP DUNILL
LAC I (ATEM+4
SAD F
JMP R2

```

```

/ILLEGAL SYNTAX DUN,DTM,DDT
DUNILL, LAC A
DAC I (FSA) /RESTORE FSA
JMP R1

```

/CURRENT YEAR

DDTCY, LAW SUTD
 PEEK
 DAC I (USERAC
 JSP .DNUM
 LAC F
 JMP DDTGY

/GET DATE

/BREAK DOWN FIELDS TO GET YEAR

/N DAYS AGO

DDTY, STF 2
 CLL"U"CML
 LAC G
 DAC I (FSA
 CLF 6
 JSP .DNM
 JMP DDTG

/FLAG 2 MAKES .DNM COMP. #
 /LINK SET MEANS "Y" AFTER .DNM
 /N DAYS HENCE
 /STEP PAST Y OR T
 /NO MAX

DDTOI, LAW SUTD

 PEEK
 ADD I (USERAC
 DAC I (USERAC
 JMP R2

/GET DATE
 /ADD N

/NO NUMBER AFTER Y OR T

```
DDTO,      LAC I (USERAC) /FIND OUT WHY .DNM FAILED
           SZA I          /IF OVERFLOW, FAIL
           SZL
           JMP DUNILL
           JMP DDTOI      /AC=0, TODAY
```

```
DCDTD,     LIO (USERAC) /PTR, TO BLOCK IN AC
           JSP TRACEI
           LAW 24.       /CHECK THAT HOUR IS LEGAL
           SUB I B
           LIO I B
           DIO A         /SAVE UP HOUR
           IAI
           SPA
           JMP R1        / "I B" >23, OR <0
           IDX B
           LAW 60.       /CHECK THAT MINUTE IS LEGAL
           SUB I B
           IOR I B
           SPA
           JMP R1        / "I B" >59, OR <0
           LAW 60.
```

```
CHARLL,    MUL A
           SCL 9S
           SCL 8S
           ADD I B
           DAC I (USERIO) /COMPLETE TIME
           IDX B
           LAC I B
           DAC I (ATEM+2)
           IDX B
           LAC I B
           DAC I (ATEM+3)
           IDX B
           LAC I B
           DAC I (ATEM+4)
           LAC I (FSA)
           DAC A         /PUT FSA IN A SO DUNILL DOES'NT KILL FSA
           JMP DCDTDI
```

/TT TRAP CONDITION ON TYPEOUT SETUP ROUTINES

```
SICKTT,    LAC I (USERPC)
           DAC I (TRAPPC)
           LAW TTTSU
```

/RETURN ENTRIES TO USER CORE

```
GO,        DAP I (USERPC) /GO THRU AC
IR0,       C16RET        /INDEXABLE RETURN 0
IR1,       JMP R1        /INDEXABLE RETURN 1
R2,        IDX I (USERPC) /RETURN 2
R1,        IDX I (USERPC) /RETURN 1
           C16RET        /TO USER CORE
```

START